

Multimodal Deep Learning Search Engine for E-commerce Fashion

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Problem & motivation

- Shoppers describe products in natural language - e.g. *"a flowy floral dress with cap sleeves"* - but catalogs rely on keywords.
- We build a retriever that maps **text queries** -> **product images** in a shared embedding space.
- Dataset: **44,265** fashion products (Kaggle Fashion Product Images).

Research question

*Can a dual-encoder deep learning model, trained on paired fashion images and text, **outperform traditional keyword-based e-commerce search**?*

- **Baseline:** TF-IDF on product_text (caption + JSON description).
- **Proposed:** Dual-encoder - text query -> embedding <- product image embedding.

Dataset & preprocessing

Item	Detail
Source	Kaggle Fashion Product Images (~44k)
Splits	80/10/10 train / val / test
Test gallery	4,427 products

Pipeline: `src/prepare_data.py`

Query styles

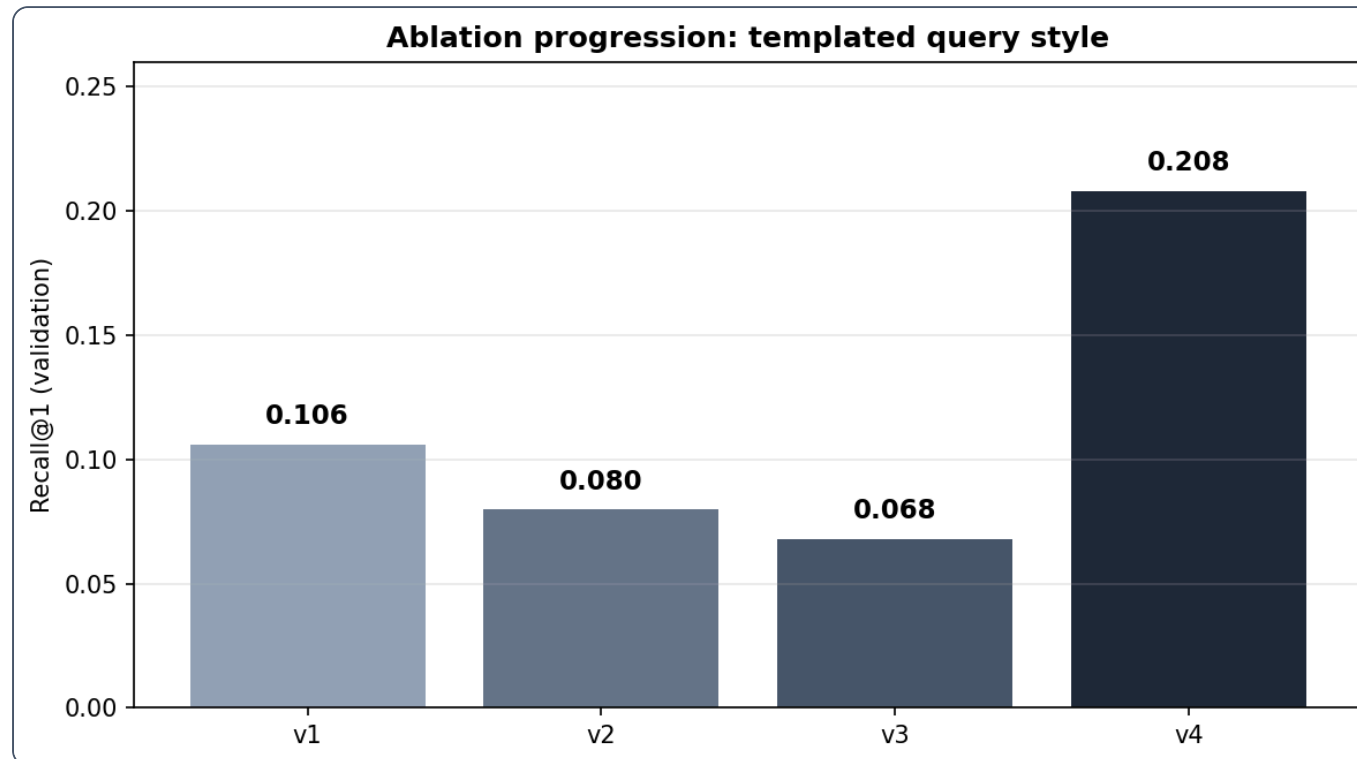
Style	Example
templated	A men navy blue shirts, for casual wear in fall...
shopper	men's navy blue shirt
brand	Turtle Check Men Navy Blue Shirt
short	navy shirt

Architecture (v4)

```
Text query -> MiniLM (frozen) -> 384-d embedding
                                   cosine similarity
Product image -> EfficientNetB0 (fine-tuned) -> 384-d embedding
```

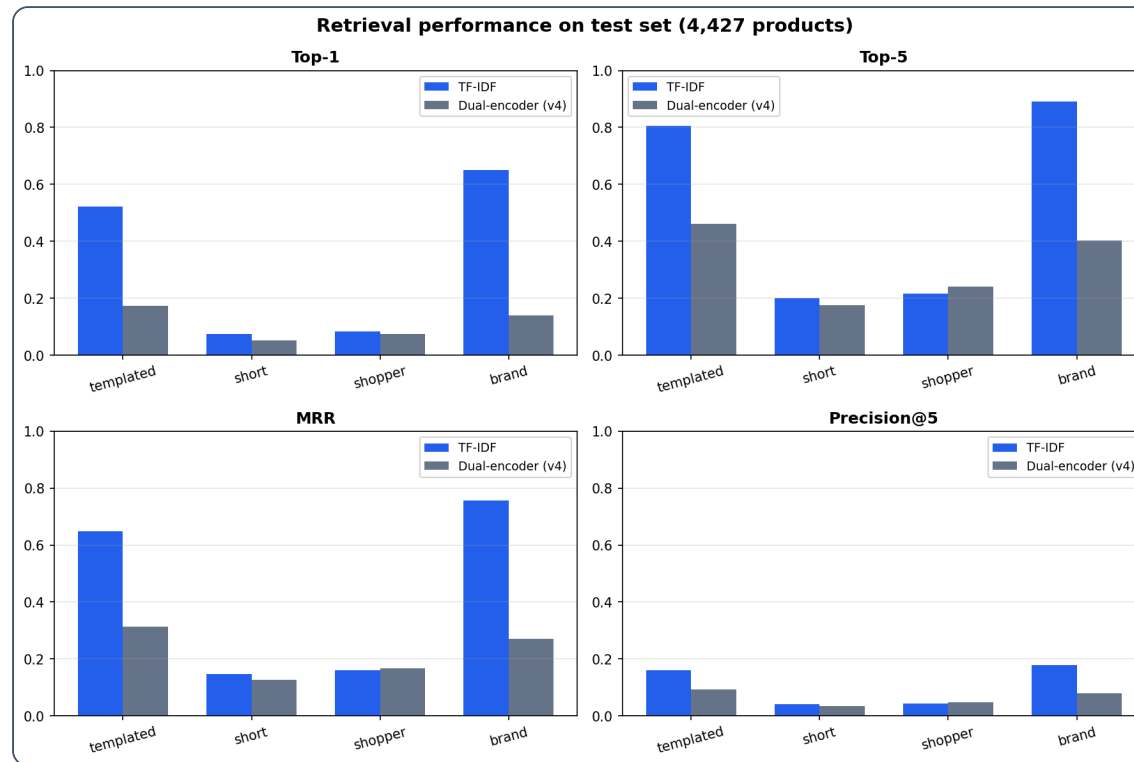
- Contrastive loss, $\tau = 0.07$
- Selected after v1->v5 ablations - pretrained text encoder = largest gain

Ablation progression (validation)



Recall@1 on templated queries - v4 (MiniLM) nearly doubled v1

Results - test set (4,427 products)

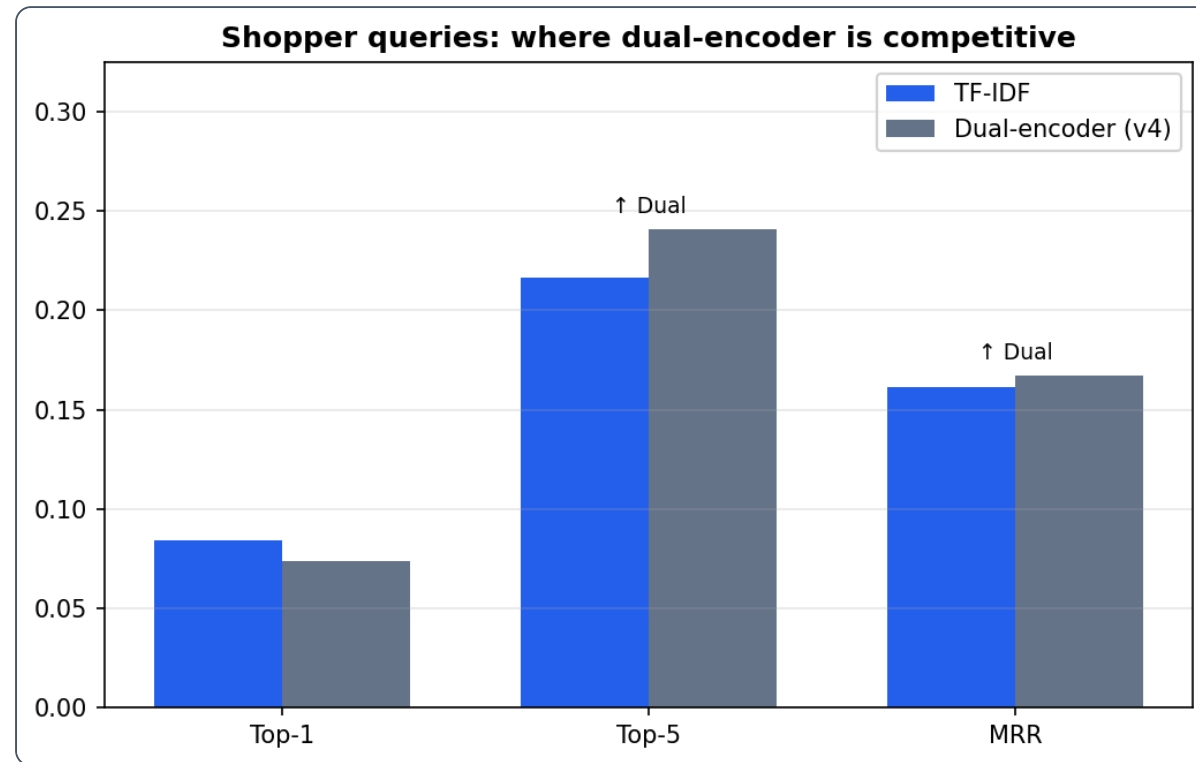


TF-IDF wins Top-1 on every query style

Top-1 accuracy

Query type	TF-IDF	Dual-encoder
templated	0.52	0.17
shopper	0.08	0.07
brand	0.65	0.14
short	0.07	0.05

Where dual-encoder wins (shopper)



Higher Top-5 (0.24 vs 0.22) and MRR (0.167 vs 0.161)

Demo - success (templated query)

Success — dual-encoder (templated query)

Query: A unisex black backpacks, for casual wear in fall. Timberland Unisex Solid Black Backpacks.
Correct product: rank 1

Timberland Unisex Solid Black
(correct)



Nike Unisex Brasilia 4 XL Ba



Nike Unisex Football Black B



ADIDAS Unisex Navy Blue Casu



ADIDAS Unisex Bess Bp Black



Demo - head-to-head (brand query)

Head-to-head: same brand query Baggit Women Teal Wallet

Baggit Women Teal Wallet



Fabindia Men Teal Tussar



Rocia Women Teal Heels



Cobblerz Men Teal Blue C



Kiara Women Teal Handbag



Baggit Women Grey Tijori



Baggit Women Highway Taj



Baggit Women Voodoo Viva



Baggit Women Auxi Baby P



Baggit Women Red Famous



Demo - failure (shopper query)

Failure — shopper query (dual-encoder)
Query: women's red handbag
Correct product: not in top-5

Kiara Women Black Handbag



Lino Perros Women Red Handba



Hidekraft Women Red Handbag



Nyk Women Red Pacifier Handb



Murcia Women Red Handbag



Limitations

- **Modality gap** - text->image vs TF-IDF text->text
- **Low Top-1** - both models struggle on shopper/short (~5-8%)
- **Synthetic queries** - not real search logs
- **Compute** - ~40 min CPU training

Conclusion & future work

Conclusion: Keyword search wins Top-1 when the gallery is text-indexed. Dual-encoder is viable for cross-modal retrieval and wins on shopper Top-5 / MRR.

Future work: fine-tune text tower, hard negatives, CLIP-style pretraining, hybrid TF-IDF -> dual reranking.

Full report: docs/reports/final_report.pdf - Repro: GRADING.md

Thank you!